AuraDB Knowledge Graph Creation

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2.Create a schema and import the data into AuraDB.

Data source **Local** Browse ···

course+learning_outcome.csv ···

course_id	DE0745
outcome_id	CLO_49b5d0eaac4149

course+topic.csv ···

course_id	DE0745
topic_id	TOPIC_1ea7cdfb55c5ed

nodes_course_topics.csv ···

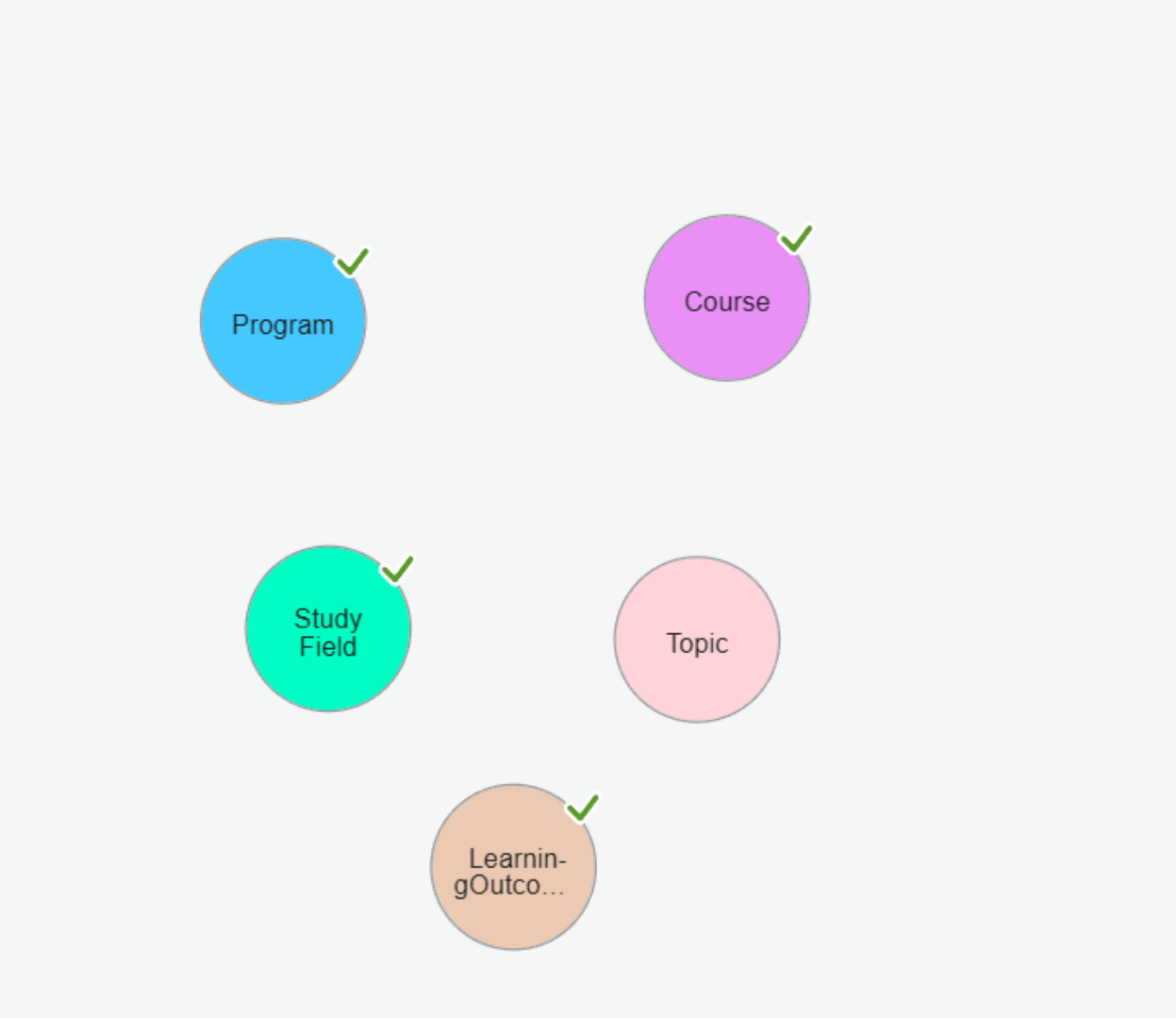
topic_id	TOPIC_1ea7cdfb55c5ed
course_id	DE0745
title	Universal database systems: relati...
ordinal	1
intramural_contact_hours	8
intramural_independent_hours	10
extramural_contact_hours	0
extramural_independent_hours	0

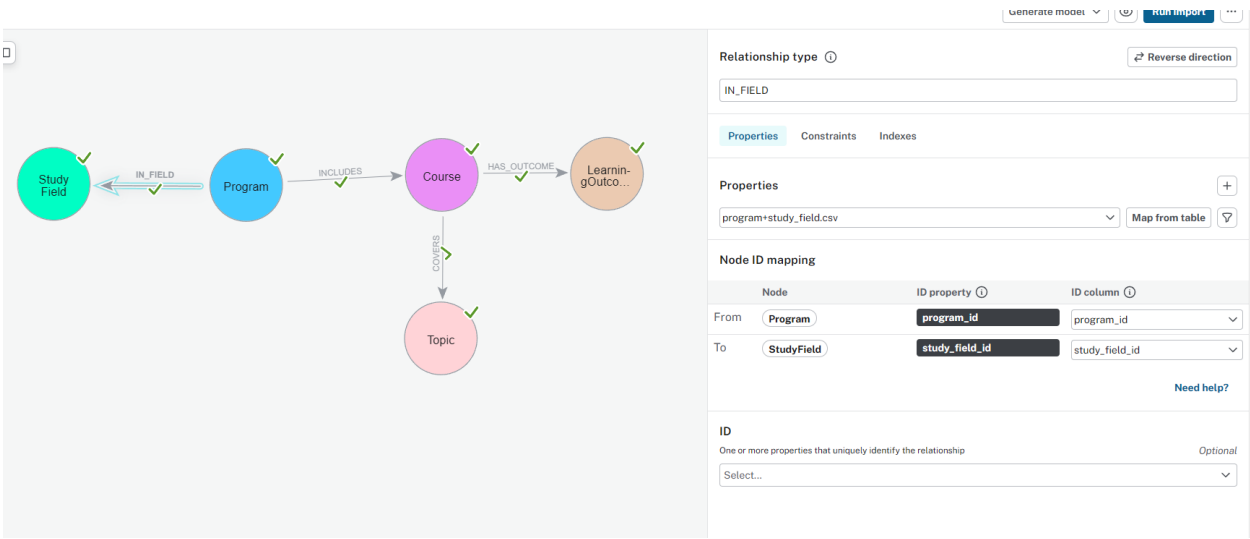
nodes_learning_outcomes.csv ···

outcome_id	PLO_1ebb9af2c72835
text	can identify business goals which ...
scope	programme
owner_id	DMB0
ordinal	1

nodes_study_courses.csv ···

course_id	DE0745
title	Advanced data technologies
status_in_programme	Compulsory/Courses of Limited C...
credit_points	6.0
languages	LV, EN
abstract	Adequate technology is needed to ...
aims	The aim of the study course is to a...
independent_study	The knowledge gained in lectures i...
recommended_literature	1. Obligātā / Obligatory 2. Data C...


Program
Course
Study Field
Topic
Learnin-gOutco...


Study Field
IN_FIELD
Program
INCLUDES
Course
HAS_OUTCOME
Learnin-gOutco...
COVERS
Topic
Generate model
Run import
Relationship type
Reverse direction
IN_FIELD
Properties
Constraints
Indexes
Properties
program+study_field.csv
Map from table
Node ID mapping
Node
ID property
ID column
From
Program
program_id
program_id
To
StudyField
study_field_id
study_field_id
Need help?
ID
One or more properties that uniquely identify the relationship
Optional
Select...

Import results

Total time: 00:00:01

Run import completed

Import completed successfully.

Program nodes_study_programs.csv — Show Cypher

Time taken	File size	File rows	Nodes created	Properties set	Labels added	Query time
00:00:00	13.3 KiB	2	2	8	2	00:00:00

StudyField nodes_study_fields.csv — Show Cypher

Time taken	File size	File rows	Nodes created	Properties set	Labels added	Query time
00:00:00	165 B	1	1	1	1	00:00:00

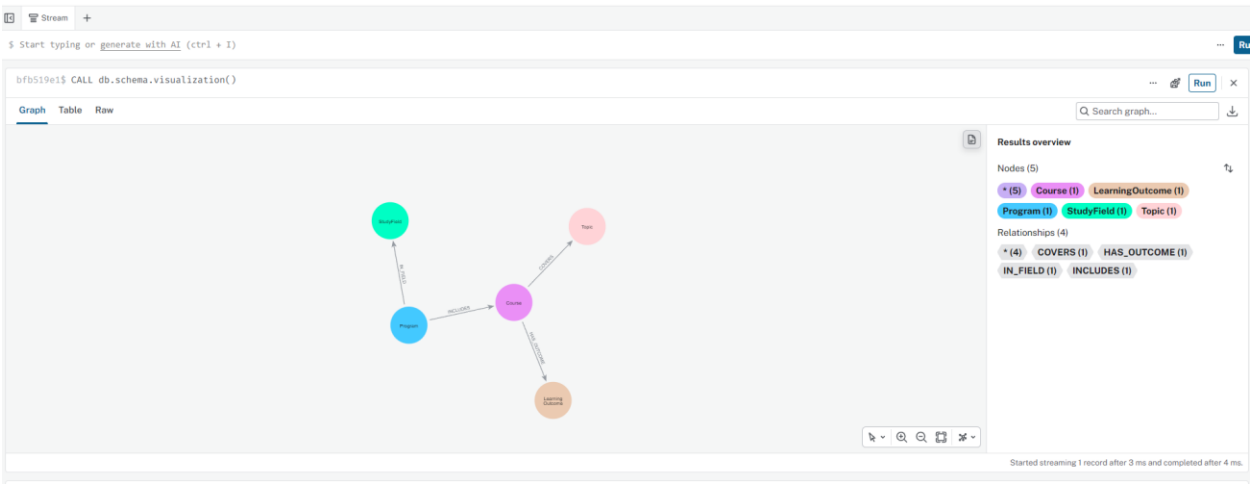
Course nodes_study_courses.csv — Show Cypher

Time taken	File size	File rows	Nodes created	Properties set	Labels added	Query time
00:00:00	284.0 KiB	75	75	299	75	00:00:00

Topic nodes_course_topics.csv — Show Cypher

Time taken	File size	File rows	Nodes created	Properties set	Labels added	Query time

Close | Explore with agent



3.Run three to five queries on the created knowledge graph.

1 QUERY:

MATCH (P:Program)-[:INCLUDES]->(C:Course)

WITH C, collect(P.title) AS programmes

WHERE size(programmes) > 1

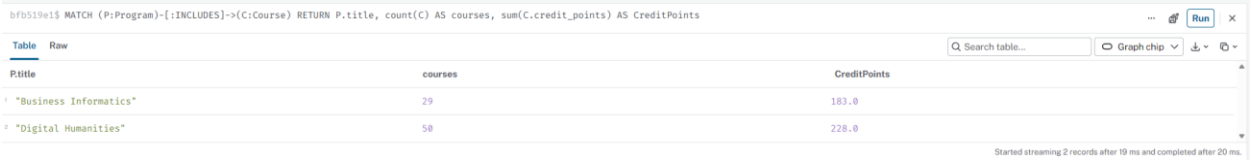
RETURN C.course_id, C.title, programmes



2QUERY:

MATCH (P:Program)-[:INCLUDES]->(C:Course)

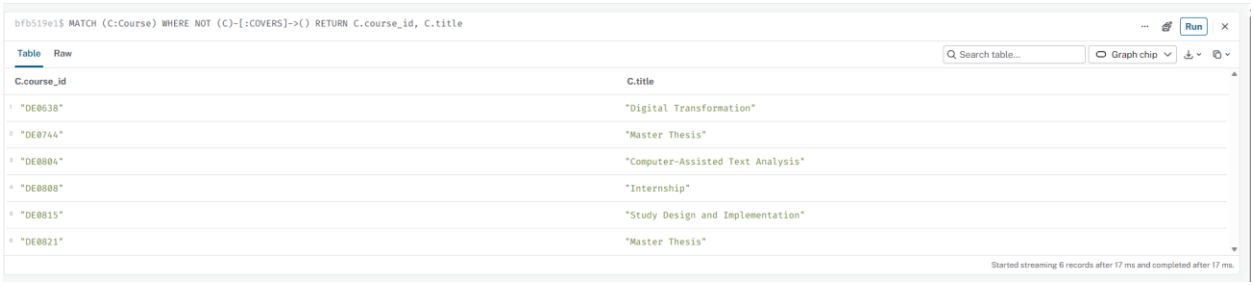
RETURN P.title, count(C) AS courses, sum(C.credit_points) AS CreditPoints



3QUERY:

MATCH (C:Course) WHERE NOT (C)-[:COVERS]->()

RETURN C.course_id, C.title



4QUERY:

MATCH (P:Program)-[:INCLUDES]->(C:Course)-[:COVERS]->(T:Topic)

WHERE toLower(T.title) CONTAINS 'data'

RETURN C.title, collect(DISTINCT P.program_id) AS Programme, count(DISTINCT T) AS topics

ORDER BY topics DESC

